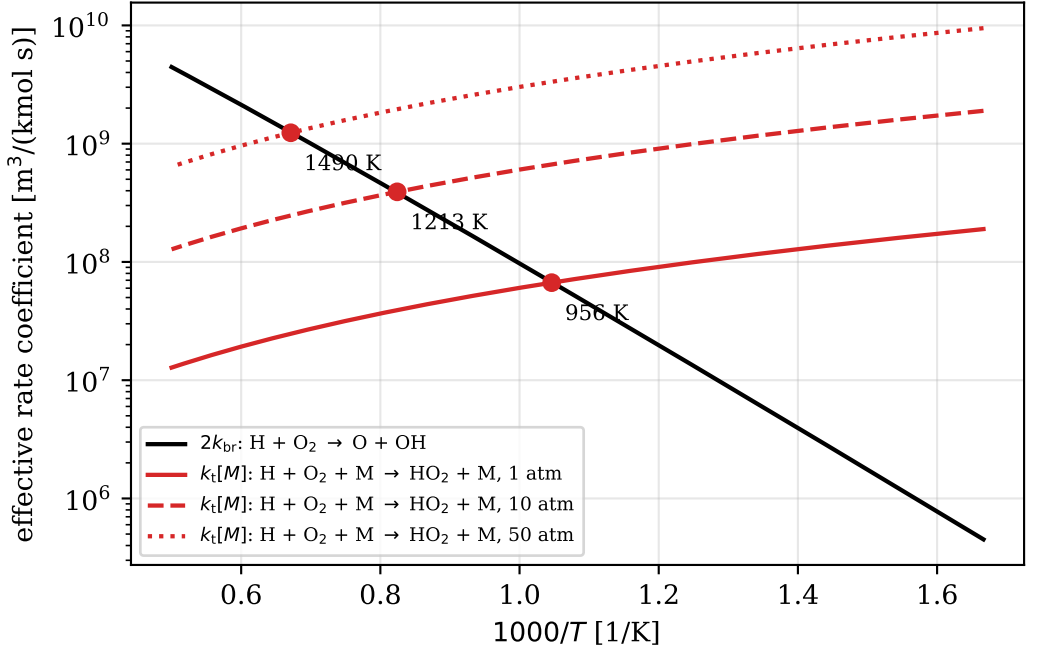
Chain branching vs. termination competing for H atoms
effective rate coefficient [m^3/(kmol s)]
10^{10}
10^9
10^8
10^7
10^6
1490 K
1213 K
956 K
$2k_{br}$: H + O_2 → O + OH
$k_t[M]$: H + O_2 + M → HO_2 + M, 1 atm
$k_t[M]$: H + O_2 + M → HO_2 + M, 10 atm
$k_t[M]$: H + O_2 + M → HO_2 + M, 50 atm
0.6
0.8
1.0
1.2
1.4
1.6
1000/T [1/K]